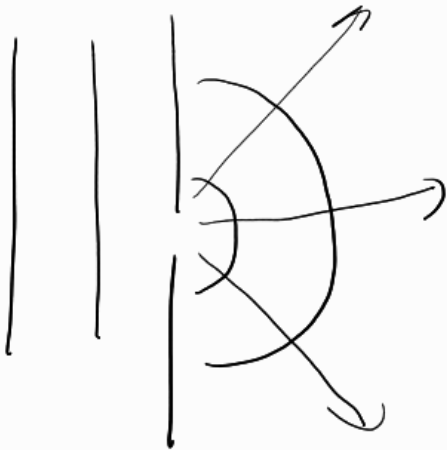


- drevo ustavi svetlobo, ne pa npr. radijskih valov

→ uklon

$$(d \ll \lambda)$$

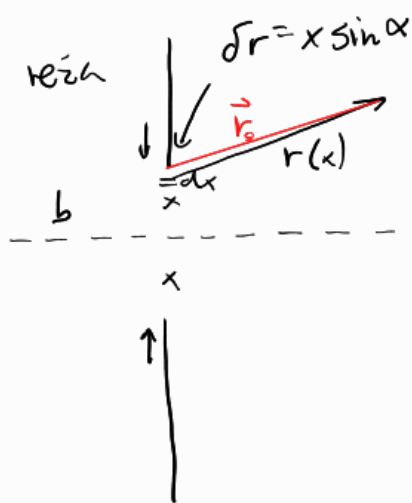


En tip na H je rekni:

- vsaka točka val. fronte je izvir kroglnega val. v smeri naprej
- celotno valovanje je Σ teh kroglnih valov

- interferenca (Youngov poskus)

$$a \sin \alpha = N \lambda$$



$$\frac{b}{2} \sin \alpha = \frac{\lambda}{2}; \text{ min}$$

$$b \sin \alpha = \lambda$$

$$dE(x) = dE_0(r) \cos(kr - \omega t)$$

Frennhofer → gledano daleč stran

$$E_0 \neq E_0(r)$$

$$dE_0 = \frac{dx}{b} E_0$$

$$dE(x) = \frac{E_0}{b} dx \cos(k(r_0 + x \sin \alpha) - \omega t)$$

$$E(x) = \int_0^b \cos(k(r_0 + x \sin \alpha) - \omega t) dx$$

→ gremo v $\mathbb{C}$

$$A \cos(kr - \omega t) \rightarrow A e^{i(kr - \omega t)}$$

$$j \propto E_0^2 \rightarrow j \propto E^* E$$

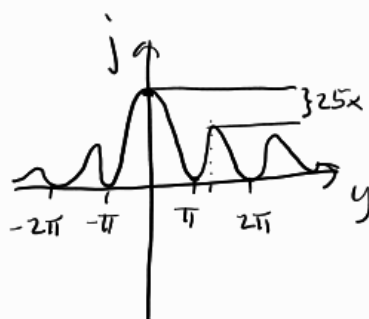
$$\begin{aligned} \mathcal{E} &= \frac{E_0}{b} \int_0^b dx e^{i(kr_0 + kx \sin \alpha - \omega t)} = \frac{E_0}{b} e^{i(kr_0 - \omega t)} \int_0^b e^{ikx \sin \alpha} dx = \\ &= \frac{E_0}{b} e^{i(kr_0 - \omega t)} \frac{e^{ikx \sin \alpha}}{ik \sin \alpha} \Big|_0^b = \frac{E_0}{ik \sin \alpha} e^{i(kr_0 - \omega t)} [e^{ikb \sin \alpha} - 1] = \\ &= \frac{E_0}{b k \sin \alpha} e^{i(kr_0 - \omega t)} e^{i \frac{kb}{2} \sin \alpha} \frac{2}{2i} \left[e^{i \frac{kb}{2} \sin \alpha} - e^{-i \frac{kb}{2} \sin \alpha} \right] = \end{aligned}$$

$$= \frac{\sin\left(\frac{kb}{2} \sin \alpha\right)}{\frac{kb}{2} \sin \alpha} E_0$$

$$\frac{kb}{2} \sin \alpha = \pi \frac{b}{\lambda} \sin \alpha = x$$

$$j(\alpha) \propto \frac{\sin^2 x}{x^2} = E_0^2 \frac{\sin^2 x}{x^2}$$

$$j = j_0 \frac{\sin^2 x}{x^2}$$

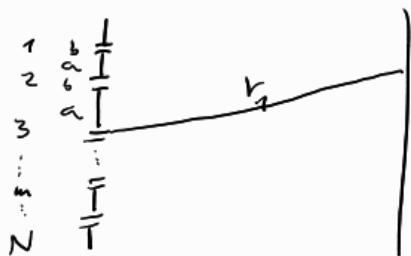


ničle: $y = N\pi$

$$\frac{kb}{2} \sin \alpha = N\pi$$

$$b \sin \alpha = N\lambda$$

N -rež, širina reže b , razd. med režama a



$$\mathcal{E} = \sum_{m=1}^N \mathcal{E}_m = \sum E_0 \cos(kr - \omega t) = E_0 \sum_{m=1}^N e^{i(kr_m + k(m-1)a \sin \alpha - \omega t)}$$

$$= E_0 e^{i(kr_1 - \omega t)} \sum_{m=1}^N e^{i(m-1)a \sin \alpha} = E_0 e^{i(kr_1 - \omega t)} [1 + e^{i\delta} + e^{2i\delta} + \dots + e^{i(N-1)\delta}] = \frac{e^{iN\delta} - 1}{e^{i\delta} - 1} = \frac{\sin\left(\frac{N\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)}$$

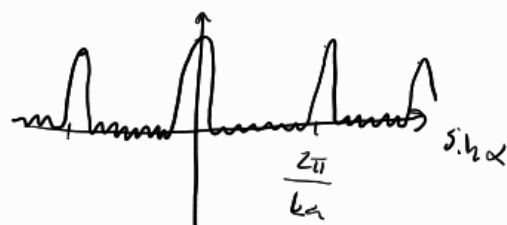
$$\rightarrow \mathcal{E} = E_0 \frac{\sin\left(\frac{N\delta}{2}\right)}{\sin\left(\frac{\delta}{2}\right)} \quad \delta = ka \sin \alpha$$

$$j(\alpha) = j_0 \frac{\sin^2 N\delta}{\sin^2 \delta} ; \quad y = \frac{ka}{2} \sin \alpha$$

$$a \sin \alpha = n\lambda$$

$$j(0) = N^2 j_0$$

na simetriji je močno ojačanje



Več kot je rež, bolj so ostri vrhovi



$$\boxed{\frac{\sin \alpha}{\sin \beta} = \frac{c_1}{c_2} = \frac{n_2}{n_1}}$$

lomni zakon

$$n = \frac{c_0}{c}$$

$$\boxed{\alpha' = \alpha}$$

odbojni zakon

N-širokih rež: ($b < a$)

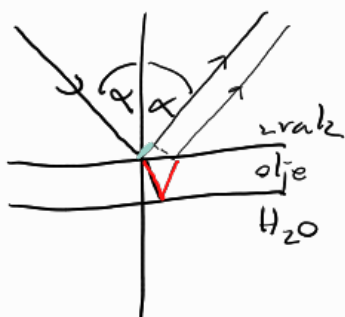
$$j = j_0 \underbrace{\frac{\sin^2 N y}{\sin^2 y}}_{\text{strukturni faktor}} \cdot \underbrace{\frac{\sin^2 x}{x^2}}_{\text{režni faktor}}$$

Vklon na tankih plasteh

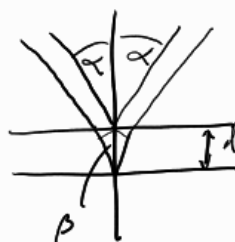
$$c = \lambda' \nu = \frac{c_0}{n} \rightarrow n \lambda' = \lambda_0$$

λ_0 v vakuumu

faza v shoji! $\delta = k d = \frac{2\pi}{\lambda'} d = \frac{2\pi}{\lambda_0} \underbrace{(n d)}_{\text{optična pot}} = \underline{k_0 (n d)}$



vs. $\Rightarrow$



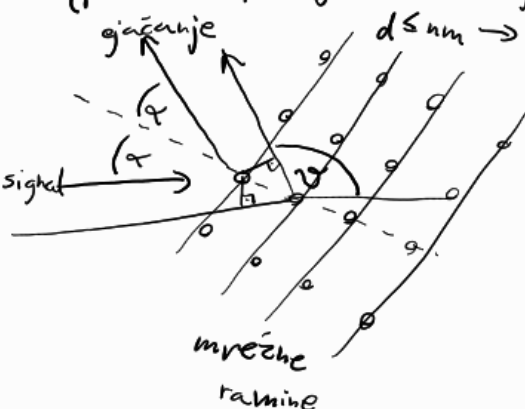
$$2 n d \cos \beta = N \lambda \quad \text{min}$$

$$\searrow N(\lambda + \frac{\lambda}{2}) \quad \text{max}$$

obratno kot pričakovali, ker se pri odboju $n_1 < n_2$ obrne faza.

Interferenca na kristalu

(poskus - spenjanje na odboj)



$$d \leq n \lambda \rightarrow \lambda \geq d \rightarrow \text{rtog. žarki} \rightarrow \underline{n=1}$$

$$2 d \cos \alpha = N \lambda$$

$$2 \alpha + \nu = \pi$$

$$\Rightarrow \boxed{2 d \sin \frac{\nu}{2} = N \lambda}$$

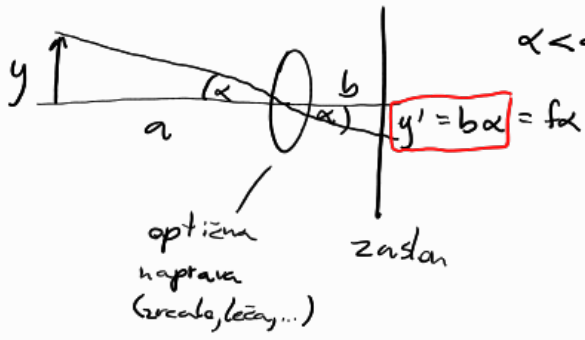
Braggova enačba

(določanje strukture in razdalj v kristalu)

Geom. optika

$\alpha \rightarrow$ zorní kot

$\alpha \ll$



rozdílnost α_{min}

$$\alpha_{min} = \frac{x}{b}$$

M-povečava

zrak oko: $\alpha_{min} \approx 1''$

